A3 Solutions

1. (a) $\lim_{x\to 3}(x-1)^2(x+1) = (3-1)^2(3+1) = 16$

(b) $\lim_{x\to 5}\frac{x+3}{5-x}$ doesn't exist.

Because $\lim_{x\to 5} x+3 = 8$, the limit of the denominator $\lim_{x\to 5} 5-x = 0$.

(c) $\lim_{x\to 3}\frac{9-x^2}{x-3} = \lim_{x\to 3}\frac{(3-x)(3+x)}{x-3} = -(3+3) = -6$

(d) $\lim_{x\to 5}\frac{x^2-3x-10}{x-5} = \lim_{x\to 5}\frac{(x-5)(x+2)}{x-5} = \lim_{x\to 5} x+2 = 7$

(e) $\lim_{x\to -2}\frac{x^2-x-6}{x^2+3x+2} = \lim_{x\to -2}\frac{(x+2)(x-3)}{(x+2)(x+1)} = \lim_{x\to -2}\frac{x-3}{x+1} = 5$

(f) $\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4} = \lim_{x\to 4}\frac{\sqrt{x}-2}{(\sqrt{x}-2)(\sqrt{x}+2)} = \lim_{x\to 4}\frac{1}{\sqrt{x}+2} = \frac{1}{4}$

(g) $\lim_{x\to\infty}(\sqrt{x^2+3}-x) = \lim_{x\to\infty}\frac{\left(\sqrt{x^2+3}\right)^2 - x^2}{\sqrt{x^2+3}+x} = \lim_{x\to\infty}\frac{3}{\sqrt{x^2+3}+x} = 0$

(h) $\lim_{x\to-\infty}\left(1+\frac{3}{x}\right)^{2x}$

Firstly let's prove that $\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x = e$. (using $\lim_{n\to+\infty}\left(1+\frac{1}{n}\right)^n = e$)

When $x>1$, we have $1+\frac{1}{[x]+1} \le 1+\frac{1}{x} \le 1+\frac{1}{[x]}$

$$\Rightarrow \left(1+\frac{1}{[x]+1}\right)^x \le \left(1+\frac{1}{x}\right)^x \le \left(1+\frac{1}{[x]}\right)^x$$

$$\Rightarrow \left(1+\frac{1}{[x]+1}\right)^{[x]} \le \left(1+\frac{1}{x}\right)^x \le \left(1+\frac{1}{[x]}\right)^{[x]+1}$$

$$\lim_{x\to+\infty}\left(1+\frac{1}{[x]+1}\right)^{[x]} = \lim_{x\to+\infty}\left(1+\frac{1}{[x]+1}\right)^{[x]+1}\left(1+\frac{1}{[x]+1}\right)^{-1} = e$$

$$\lim_{x\to+\infty}\left(1+\frac{1}{[x]}\right)^{[x]+1} = \lim_{x\to+\infty}\left(1+\frac{1}{[x]}\right)^{[x]}\left(1+\frac{1}{[x]}\right) = e$$

Let $y = -\frac{x}{3}$, then $y\to+\infty$ as $x\to-\infty$, and $x = -3y$.

$$\lim_{x\to-\infty}\left(1+\frac{3}{x}\right)^{2x} = \lim_{y\to+\infty}\left[\left(1+\frac{1}{y-1}\right)^{y-1}\left(1+\frac{1}{y-1}\right)\right]^6 = e^6$$

2. (a) $\lim_{x\to+\infty} x^3-4x^2-4 = +\infty$

$\lim_{x\to-\infty} x^3-4x^2-4 = -\infty$

(b) $\lim_{x\to\pm\infty} \frac{x^2-2x+3}{2x^2+5x+1} = \lim_{x\to\pm\infty} \frac{1-\frac{2}{x}+\frac{3}{x^2}}{2+\frac{5}{x}+\frac{1}{x^2}} = \frac{1}{2}$

(c) $\lim_{x\to\pm\infty} \frac{3x^2-6x+2}{2x-9} = \lim_{x\to\pm\infty} \frac{x^2}{x}\cdot\frac{3-\frac{6}{x}+\frac{2}{x^2}}{2-\frac{9}{x}} = \pm\infty$

3. (a) $\lim_{x\to3+} \frac{\sqrt{x+1}-2}{x-3} = \lim_{x\to3+} \frac{\sqrt{x+1}-2}{(\sqrt{x+1}+2)(\sqrt{x+1}-2)} = \lim_{x\to3+} \frac{1}{\sqrt{x+1}+2} = \frac{1}{4}$

(b) $\lim_{x\to3-} f(x) = \lim_{x\to3-} 2x^2-x = 15$

$\lim_{x\to3+} f(x) = \lim_{x\to3+} 3-x = 0$

4. (a) According to the definition of the limit of function at c:
For every $\varepsilon>0$, there exists a $\delta>0$ such that $0<|x-c|<\delta$. implies $|f(x)-L|<\varepsilon$.
So, let's take any sequence $x_n \overset{n\to\infty}{\to} c$ but $x_n \neq c$. Then for the δ above, there exists an N such that for all $n>N$, $|x_n-c|<\delta$. That implies $|f(x_n)-L|<\varepsilon$. This is precisely the definition of the limit of a sequence, $f(x_n) \overset{n\to\infty}{\longrightarrow} L$.

(b). We'll prove this by contradition. Suppose that $\lim_{x\to c} f(x) \neq L$. Then, there exists an $\varepsilon>0$ such that for all $\delta>0$, there exists an x with $0<|x-c|<\delta$ and $|f(x)-L|>\varepsilon$.
Now, for each $n\in\mathbb{N}$, choose x_n such that $0<|x_n-c|<\frac{1}{n}$. and $|f(x_n)-L|>\varepsilon$. Then, $\{x_n\}$ is a sequence converging to c, but $\lim_{n\to\infty} f(x_n) \neq L$, which contradicts the assumption.
Therefore, $\lim_{x\to c} f(x) = L$.

5. Proof:

(a) By definition, for every $\varepsilon>0$, there exists a $\delta>0$ such that if $0<|x-a|<\delta$, then $|f(x)-L|<\varepsilon$.
Here, $f(x)=c$. So, $|f(x)-L|=|c-c|=0<\varepsilon$ for all x and for all $\varepsilon>0$. Therefore, $\lim_{x\to a} c=c$.

(b) We need to show that for every $\varepsilon>0$, there exists a $\delta>0$ such that if $0<|x-2|<\delta$, then $|x^3-8|<\varepsilon$.
If we choose $\delta=\min(1,\frac{\varepsilon}{20})$, then for $0<|x-2|<\delta$, we have $|x^3-8|=|x-2||x^2+2x+4|<\delta[(2+\delta)^2+2(2+\delta)+4]<\varepsilon$.
Therefore $\lim_{x\to 2} x^3=8$.

(c) If we choose $\delta=\varepsilon$, then for $0<|x-2|<\delta$, we have $|\sqrt{x}-\sqrt{2}|=\frac{|x-2|}{\sqrt{x}+\sqrt{2}}<\frac{|x-2|}{\sqrt{2}}<|x-2|<\delta=\varepsilon$.
Therefore $\lim_{x\to 2}\sqrt{x}=\sqrt{2}$.

(d) The definition of the limit being infinity is that for every $M>0$, there exists a $\delta>0$ such that if $0<|x-(-3)|<\delta$, then $f(x)>M$.
If we choose $\delta=\frac{1}{\sqrt[4]{M}}$, then for $0<|x-(-3)|<\delta$, we have $f(x)=\frac{1}{(x+3)^4}>\frac{1}{\delta^4}=M$.
Therefore $\lim_{x\to -3}\frac{1}{(x+3)^4}=+\infty$.

6. Since $|\sin x|\le|x|$ for all x, we can choose $\delta=\varepsilon$. Then, for $0<|x|<\delta$, we have $|\sin x|\le|x|<\delta=\varepsilon$.
Therefore, $\lim_{x\to 0}\sin x=0$.
We know that $|\cos x-1|=2\sin^2(\frac{x}{2})\le 2|\frac{x}{2}|^2=\frac{1}{2}x^2$, So, if we choose $\delta=\sqrt{2\varepsilon}$, then for $0<|x|<\delta$, we have $|\cos x-1|\le\frac{1}{2}|x|^2<\varepsilon$. Therefore, $\lim_{x\to 0}\cos x=1$.

7. Proof: Since $f(x)$ is continuous at $x=0$, $\lim_{x\to 0} f(x) = f(0)$.
The limit of a product is the product of the limits, as long as both limits exist. We know that $\lim_{x\to 0} x = 0$.
Therefore, $\lim_{x\to 0} x f(x) = 0 \cdot f(0) = 0$.

8. Proof: (Use the conclusion in Problem 4)
Choose any sequence of rational numbers that converges to 0, say $x_n = \frac{1}{n}$, Then $f(x_n) = 0$ for all n, so $\lim_{n\to\infty} f(x_n) = 0$.
Choose any sequence of irrational number that converges to 0, say $x_n = \frac{\sqrt{2}}{n}$, Then $f(x_n) = 1$ for all n, so $\lim_{n\to\infty} f(x_n) = 1$.
Since these two limits are not equal, $\lim_{x\to 0} f(x)$ does not exist.

9. Proof:
We define the function $f(x) = \sqrt[3]{x-8} + 9x^{2/3} - 29$
We can see that $f(x)$ is continuous on the interval $[0, 8]$.
We have $f(0) = -31$ and $f(8) = 7$.
Since $-31 < 0 < 7$, by the Intermediate Value Theorem, there exists at least one $c \in (0, 8)$ such that $f(c) = 0$.
This means that the equation $\sqrt[3]{x-8} + 9x^{2/3} = 29$ has at least one solution for the interval $0 \le x \le 8$.